

UNIVERSITI TUNKU ABDUL RAHMAN

ACADEMIC YEAR 2025/2026

DECEMBER EXAMINATION

UCCD1213/UCCG1213 FUNDAMENTALS OF DIGITAL MEDIA TECHNOLOGY
UCCG1113 INTRODUCTION TO MULTIMEDIA

FRIDAY, 2 JANUARY 2026

TIME : 9.00 AM – 11.00AM (2 HOURS)

EXAMINATION FOR BACHELOR DEGREE

Instruction to Candidates :

This question paper consists of **THREE (3)** questions in **Section A** and **TWO (2)** questions in **Section B**.

Answer **ALL** questions in **Section A** and **ONLY ONE (1)** question in **Section B**. Each question carries 25 marks.

Should a candidate answer more than **ONE (1)** question in **Section B**, marks will only be awarded for the **FIRST** question in that section in the order the candidate submits the answers.

Candidates are not allowed to use calculator.

Answer questions only in the answer booklet provided.

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Section A. Answer ALL Questions. [Total: 75 marks]

- Q1. (a) Explain **TWO (2)** benefits using multimedia technology in the following industries.
- (i) Government. (4 marks)
- (ii) Business. (4 marks)
- (b) Variable-length encoding (VLE) is one of the common techniques for data compression. Perform a VLE based on Huffman Coding on the following phrase “ABBAACDCCAEAAA” for compression. Apply the appropriate steps to show the calculation of the total number of bits required to represent all the characters in the phrase. (9 marks)
- (c) Explain **TWO (2)** reasons why authoring tools are considered a high-level tool. (4 marks)
- (d) Provide **FOUR (4)** basic components required to digitize a video. (4 marks)
 [Total: 25 marks]
- Q2. (a) A target audience refers to the specific group of users most likely to engage with a multimedia product. Identifying the target audience is a crucial phase in multimedia authoring.
- (i) Evaluate **THREE (3)** important reasons why identifying the target audience is essential during the multimedia authoring process. (9 marks)
- (ii) Based on your answer in Q2(a)(i), provide **THREE (3)** relevant real-world examples for each reason to support your evaluation. (6 marks)
- (b) Briefly explain the purpose of authoring tools. (2 marks)
- (c) One of the deliverables during the planning phase is storyboards. Briefly explain **ANY FOUR (4)** benefits of producing storyboards. (8 marks)
 [Total: 25 marks]

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- Q3. (a) Short-form video content (such as TikTok, Instagram Reels, and YouTube Shorts) has become one of the leading trends on social media platforms. These platforms rely heavily on video-based authoring tools for creating, editing, and publishing multimedia content.
- (i) Briefly describe video-based authoring tools. (3 marks)
- (ii) Analyse the impact of short video content on social media users by providing **TWO (2)** advantages and **TWO (2)** disadvantages. (8 marks)
- (b) One of the factors that determines the success of a multimedia project is the good organisation of work. Identify **ANY FOUR (4)** reasons why effective work organisation is important in a multimedia project. (8 marks)
- (c) Evaluate **THREE (3)** key factors for making successful animations. (6 marks)
- [Total: 25 marks]

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Section B. Answer only ONE (1) Question. [Total: 25 marks]

- Q4. (a) Distinguish **FOUR (4)** different characteristics between lossless compression and lossy compression. (8 marks)
- (b) Recommend **ANY FOUR (4)** suitable file formats to store vector graphic files. (4 marks)
- (c) List **THREE (3)** ways of using digital sound in multimedia projects. (3 marks)
- (d) Identify **FOUR (4)** basic font terminologies commonly used in text design. (4 marks)
- (e) Discuss **TWO (2)** reasons why copyright plays a crucial role when designing a multimedia application. (6 marks)
- [Total: 25 marks]
- Q5. (a) Differentiate **FOUR (4)** differences between raster and vector graphics. (8 marks)
- (b) Suggest **ANY FOUR (4)** suitable file formats for storing audio files. (4 marks)
- (c) Provide **THREE (3)** software or applications that are available for students to create content for presentation. (3 marks)
- (d) Mixed Reality (MR) is a technology that blends real and virtual environments, allowing physical and digital objects to interact in real-time. MR can be experienced using devices or compatible AR/VR headsets. Provide **ANY FOUR (4)** examples of MR technologies currently used by people nowadays. (4 marks)
- (e) Interactive multimedia refers to a computer-based system that allows the user to control, combine and manipulate different types of media, such as text, sound and video. Discuss **ANY TWO (2)** key factors to consider when incorporating video into a website. How might each factor affect user experience and website performance? (6 marks)
- [Total: 25 marks]